

EXB-03 · EXAM BLUEPRINT

Cognitive levels and item types

What Recall, Application and Analysis mean here, why the examination uses four-option single-best-answer items, and what that format cannot measure

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PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

Every PCL-AI item is tagged at one of three cognitive levels — Recall, Application or Analysis — and the intended mix is application-weighted. This document defines each level with the kind of stem it produces, explains the anatomy of a four-option single-best-answer item, gives the Institute's reasons for choosing that format over the alternatives, and sets out honestly what the format cannot measure and what is done about each limitation. It also lists the item types the Institute refuses to use, and why. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Cognitive levels and item types

In one paragraph. Every PCL-AI item is tagged at one of three cognitive levels — Recall, Application or Analysis — and the intended mix is application-weighted. This document defines each level with the kind of stem it produces, explains the anatomy of a four-option single-best-answer item, gives the Institute's reasons for choosing that format over the alternatives, and sets out honestly what the format cannot measure and what is done about each limitation. It also lists the item types the Institute refuses to use, and why. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates who want to know what kind of thinking the examination rewards; item writers and reviewers working to the Institute's specification; and assessment professionals or academics auditing the design.

— 1. The level matters more than the topic

Two items can sample the same knowledge area and measure entirely different things.

Which formula gives the estimate at completion when past cost performance is expected to continue?

Cost inefficiency on this project is systemic and the schedule driver has been removed. Which estimate at completion is the most defensible forecast, and why?

Both sit in knowledge area 6.3. The first asks whether a candidate has memorised a relationship. The second asks whether they can read a situation, choose between defensible methods, and justify the choice. A credential built mainly from the first kind certifies revision. A credential built mainly from the second kind certifies judgement.

This is why every item carries a cognitive level as well as a domain and knowledge-area tag, and why form assembly is constrained on both. Content sampling alone — the subject of [EXB-01 – Examination blueprint – PCL-AI §5](#) — guarantees that the right topics appear. Only the level constraint guarantees that they appear at the right depth.

— 2. The three levels

2.1 Recall

What it asks. State or recognise a definition, a rule, a relationship or what a standard requires.

What it proves. That the candidate holds precise knowledge that decisions depend on. Recall is not a lesser skill; it is a prerequisite one. A professional who thinks contingency reserve and management reserve are the same thing will misreport a baseline change as a variance, and no amount of analytical ability repairs that.

Typical stem shapes. Which statement about X is correct? · Under this standard, revenue is recognised when... · The performance measurement baseline is revised...

Where the Institute uses it. Where precision genuinely changes a decision: definitional boundaries that practitioners routinely blur, recognition tests, the distinction between two adjacent concepts. Not as filler, and not for vocabulary that a professional would reasonably look up.

2.2 Application

What it asks. Use a known method on given data — compute a figure, classify a transaction, apply a rule to a stated situation.

What it proves. That the candidate can convert knowledge into a number or a classification that someone else could rely on. This is the level at which most controls work actually happens, and it is where a candidate's understanding either survives contact with data or does not.

Typical stem shapes. Given these figures, what is...? · At cut-off, what entry does the cost report require? · Applying the published threshold, which control accounts are flagged?

Where the Institute uses it. Throughout, and especially in the quantitative knowledge areas — earned value and its indices, the estimate-at-completion family, variance decomposition, progress valuation and payment, expected value, agile forecasting. Application items supply the data and never the method; the formula policy in [EXB-01](#) §7.2 exists precisely so that selecting the method remains part of the assessment.

2.3 Analysis

What it asks. Interpret a situation in which more than one course of action is arguable, weigh the competing considerations, and select what a competent professional would defensibly do next.

What it proves. Professional judgement under incomplete information — which is what a Leader-level credential is for. An Analysis item usually contains no arithmetic error to spot. It contains a position, and the question is what you do about it.

Typical stem shapes. Given this position, what should the controls professional do next? · The most accurate statement of the position is... · The most defensible response is...

Where the Institute uses it. Wherever the discipline's real difficulty lies: choosing between two legitimate forecasting methods on the evidence, deciding whether an instructed variation is a cost or a trend, judging whether a machine-generated output can be relied on, deciding what to escalate and when.

What an Analysis item never rewards. The heroic answer. The best option is what a practitioner could defensibly do next — often verifying a figure, correcting a baseline, or telling someone something they do not want to hear — not a dramatic intervention that no controls professional has the authority to make.

— 3. The intended mix

The examination is **application-weighted**. The majority of scored items sit at Application and Analysis; Recall is the minority of the paper and appears where precise knowledge genuinely matters.

That is a statement about design intent, and it is as far as the Institute will go today. The exact proportions form part of the examination specification and will be confirmed through item development and the job-task analysis: **[CONFIRM: target proportion of items at each cognitive level, to be fixed in the examination specification]**. Publishing a precise mix before the item bank can support it would be a number invented to look rigorous.

The honest statement a candidate can act on: **expect to be tested mainly on doing and judging**. If your preparation consists of memorising definitions and formulae, you have prepared for the smallest part of the paper.

— 4. The item format

4.1 Anatomy

Every item is a **four-option multiple-choice question with a single best answer**. It has three parts.

The stem sets out the situation and supplies every figure needed to answer it. In a scenario item the stem is drawn from real controls work — a cost position at month end, a network with a delay, a payment application, a machine-generated forecast — and contains no information the candidate is expected to supply from outside the body of knowledge.

The key is the single best answer. Not merely a correct answer: the best of the four, judged against the standard the body of knowledge teaches. Where two options are both technically permissible, the key is the one a competent professional would choose on the evidence given, and the rationale says why the other is second best rather than wrong.

The three distractors are wrong, and wrong for a reason. Every distractor in a well-written item corresponds to a specific error a real practitioner makes: an inverted ratio, a cumulative figure used where an incremental one is required, a method applied outside its assumptions, a plausible action that contradicts something stated in the stem. A distractor nobody would choose measures nothing and wastes a quarter of the item.

4.2 Scoring mechanics a candidate should know

- **Every scored item carries equal weight.** The 40/40/20 weighting is achieved through the composition of the form, not through per-item weights — see [EXB-02 – Domain weightings and the 40/40/20 rule §3](#).
- **There is no negative marking.** A wrong answer scores zero. It does not score less than zero.
- **An unanswered item scores exactly what a wrong answer scores.** There is therefore never a reason to leave an item blank. If you are out of time, answer.
- **Random guessing on a four-option item has an expected value of 25 per cent ($1 \div 4$).** That is the chance floor, and it is well below any plausible passing standard — which is one reason the standard has to be set by a documented method rather than chosen as a round number. See [EXB-05 – Standard setting: modified Angoff and the pass mark](#).

4.3 How to work an item

The discipline that answers items well is the discipline the examination is trying to measure.

1. **Establish the position before reading the options.** State in your own words what the data actually says — what the indices mean, what the dates imply, what the ledger is and is not capturing. Options read first will frame the data; data read first will filter the options.
2. **Eliminate what contradicts the stem.** The most common weak distractor is a reasonable-sounding action that ignores something the scenario has already told you.
3. **Choose the defensible action, not the impressive one.**
4. **Apply the same test to any machine output in the stem.** Ask whether it is explainable, validated and owned before anything is built on it. If the option builds on an unvalidated output, it is not the key.

— 5. Why this format

The choice of a single item type across an entire examination needs a defence, because it is also a constraint.

Scoring is objective and reproducible. A key is a key. Two markers cannot disagree, a candidate cannot be advantaged by an examiner's mood or handwriting, and a challenge to a result is answered by inspecting the item rather than by re-marking a judgement. For a certification decision that must be defensible to the candidate, the employer and any appeal, this matters more than it does in education.

It is auditable at the item level. Because every response is a discrete choice, each item generates statistics — how many candidates got it right, and whether the ones who got it right were the stronger candidates overall. That is what makes item quality measurable rather than assertable, and it is the basis of the whole review and retirement process in [EXB-04 – How items are written, reviewed and retired](#).

It samples broadly in the time available. Ninety minutes of constructed response would cover a handful of knowledge areas. The blueprint commits the examination to sampling across sixty-one, and only a compact item type makes that arithmetic work. Breadth of sampling is itself a fairness property: the narrower the sample, the more a result depends on which topics happened to appear.

It delivers and translates reliably. The examination interface is available in seven languages. A discrete-choice item can be translated and reviewed for equivalence in a way that a marked essay, judged against a rubric applied by markers working in different languages, cannot.

Scenario design recovers most of what the format appears to lose. The common objection is that multiple-choice measures recall. That is true of badly written multiple-choice. An item that presents a project position and asks what to do next is measuring judgement; the response format is just how the judgement is captured.

— 6. What this format cannot do

Every one of these is a real limitation. The Institute states them rather than waiting to be told.

It measures recognition, not generation. A candidate who can identify the correct estimate at completion among four has not demonstrated that they could build one from a blank sheet. Recognition and generation are related but not identical abilities, and the format only reaches the first.

What we do about it. Scenario stems supply raw data rather than pre-computed intermediate figures, so the candidate must generate the intermediate result before any option becomes recognisable. An item that gives you the cost performance index has tested less than one that gives you earned value and actual cost.

Options cue. Seeing four possibilities constrains and prompts thinking in a way an open question does not. Distractor sets can also leak the answer through length, specificity or grammatical fit.

What we do about it. Item-writing rules require options of comparable length and grammatical structure, prohibit the key being systematically the longest or most qualified option, and require reviewers to attempt the item with the stem covered. An item answerable from the options alone is rejected.

There is no partial credit and no visible working. A candidate who selects the right method and makes an arithmetic slip scores the same as one who selected the wrong method entirely. The format sees the answer, never the reasoning.

What we do about it. We accept it, and design around it. Distractors are built from the specific intermediate errors a candidate would make, so the pattern of a candidate's wrong answers is informative even though a single item is not. This is also why the domain breakdown on a result matters: it is the only diagnostic the format can offer.

Guessing contributes. At four options, chance alone returns 25 per cent.

What we do about it. The standard-setting method in [EXB-05](#) addresses chance directly: a judge asked what proportion of minimally competent candidates would answer an item correctly is implicitly rating against the chance floor, and an item that panellists rate near or below 25 per cent is flagged as flawed rather than merely hard.

It cannot assess a sustained piece of professional work. Building a cost breakdown structure, negotiating a variation, writing a defensible basis of estimate, presenting a bad forecast to a sponsor — none of these is reachable by any multiple-choice item, however well written.

What we do about it. Nothing, within this examination — and the Institute says so rather than implying otherwise. A PCL-AI attests assessed knowledge, applied method and professional judgement at the point of assessment. It does not attest observed workplace performance, and no examination of this type does. The competency framework in [CMP-03 – PCL-AI: the fourteen competencies](#) describes what evidence of the wider capability looks like; the examination is one input to a certification decision, not the whole of it.

— 7. Item types the Institute does not use

Each of these appears in professional examinations, and each is excluded deliberately.

Excluded type	Why
True/false	A 50 per cent chance floor, and almost no capacity to carry a scenario.
Multiple response ("select all that apply")	Scoring is contested — partial credit, all-or-nothing and negative marking all give different results from the same response — and none of the schemes is obviously fairest. The complexity buys no measurement.
Negatively worded stems ("which is NOT correct?")	They test careful reading of the question rather than knowledge of the answer, and they translate badly across seven interface languages.
"All of the above" / "None of the above"	Partial knowledge of two options resolves the item without reading the third. They also make distractor analysis meaningless.
Complex multiple choice (combinations such as "i and iii only")	Cueing is severe, difficulty comes from the combinatorics rather than the content, and the statistics are hard to interpret.
Items turning on the wording of a protected standard	Reproducing protected text is prohibited, and a candidate should be assessed on what a requirement means, not on whether they have memorised its phrasing.

— 8. How this goes wrong

Preparing at the wrong level. The commonest and most expensive candidate error. Flashcards and formula memorisation prepare for the smallest part of an application-weighted paper. Working full scenarios under time pressure prepares for the largest.

Treating "single best answer" as "only correct answer". In an Analysis item, more than one option may be defensible in some circumstance. The task is to choose the best on the evidence given. A candidate who argues with the item instead of answering it loses the mark.

Reading the options first. It feels faster and reliably produces worse answers, because the option set frames the data.

Leaving items blank. There is no negative marking. A blank is a guaranteed zero where a guess is a one-in-four chance.

Item writers letting the key grow. The most persistent defect in submitted items is a key that is longer and more qualified than the distractors, because the writer wanted to make it unambiguously right. Length is a cue, and reviewers reject on it. The worked commentary in [EXB-08 – Sample items with rationale](#) shows the correction.

Mistaking a hard item for a good one. Difficulty and quality are independent. An item is good when the candidates who answer it correctly are the candidates who are stronger overall; that is a measurable property, and it is the subject of [EXB-04](#).

— 9. Checklist

For a candidate:

- [] I have practised at least one full timed set rather than only individual items.
- [] For quantitative items, I compute the intermediate figure before looking at the options.
- [] I can state, for any item I get wrong, which specific error the distractor I chose corresponds to.

- [] I never leave an item unanswered.
- [] I can recognise an Analysis item and resist the impressive option.

For an item writer or reviewer:

- [] The item carries a domain, knowledge-area and cognitive-level tag, and the level matches what the item actually demands.
- [] The stem supplies every figure needed and no figure that is not needed, except where a redundant figure is a deliberate and explained test.
- [] The item is unanswerable from the options alone — verified by attempting it with the stem covered.
- [] Options are of comparable length and parallel grammatical structure; the key is not the longest.
- [] Every distractor maps to a named, real practitioner error, recorded in the rationale.
- [] The item uses none of the excluded types in §7.
- [] No protected standard text is reproduced anywhere in the item.

The reason to publish the limits of a format alongside its defence is that a candidate, an employer and an auditor are each entitled to know what a result does and does not evidence. A PCL-AI pass says that, on the day, against a published sampling frame, the holder could read a project position and decide what to do about it. It does not say they have been observed doing the job. Any certification body that lets those two claims blur has stopped measuring and started marketing.

— Related

- [EXB-01 – Examination blueprint – PCL-AI](#) — the sampling frame, the format specification and the formula policy
- [EXB-04 – How items are written, reviewed and retired](#) — the process that turns these rules into a bank
- [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — how the chance floor and item difficulty enter the passing standard
- [EXB-08 – Sample items with rationale](#) — the format worked in full, with every distractor explained

— Sources and standards

The cognitive-level definitions are the Institute's own, expressed in its own words and aligned to the levels used to tag the published sample material in [docs/bok/](#). The item-type exclusions in §7 reflect long-standing item-writing practice in professional certification; they are stated here as the Institute's rules rather than attributed to any single published source.

No external citation is claimed for this document.

— Status and version

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